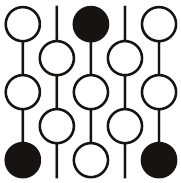
Review of Problem 7 – Astra

First Proof Third Batch
Reviewer 1
October 8–9, 2026



1 Review

Correctness.

- 1: essentially flawless mathematical argument
- 0.5: minor errors that are easily corrected
- 0: blank solution, or acknowledged partial result
- −0.5: major errors
- −1: proposed proof makes no progress towards the result.

Note that points should be docked for wrong statements/ solutions.
Score: 1

Exposition.

- 1: well-written and clear solution.
- 0: argument that is possible to follow with serious effort
- −1: very poor exposition

Score: 0

Attributions.

- 1: solutions contains fully appropriate attributions to related work and theorems being used.
- 0: moderate issues with missing attributions or hallucinated references
- −1: blatant plagiarism or hallucinations

Score: 0

> **Reviewer 1**
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> This one was easier to read than Finch and it seems that the argument only differs from Finch in a few select places. Seems correct as far as I can tell. I think this one is comparable to the human written solution.

2 Submission with inline comments

Abstract

We describe a simple closed curve for which the Dehn twist itself, not merely a proper power, preserves the deformation space in the question and has an infinite orbit on its connected components. The curve is first specified for one explicit normalized rational map and is then transported to an arbitrary map in the family. The proof uses an explicit Hurwitz covering and a free-group quotient of the stabilizer of a component. It does not assume that the components of the deformation space are simply connected.

3 The explicit curve and the statement

All marked points are labelled. Thus mapping class groups below are pure mapping class groups, and moduli spaces are moduli spaces of ordered configurations.

Lemma 3.1 (Normal form)**.** *After the unique Möbius normalization*

$$c_1 = 0, \qquad f(c_1) = \infty, \qquad f^{\circ 2}(c_1) = 1,$$

the map in the question is

$$F_r(z) = 1 - \frac{r^2 + r - 1}{(r-1)z} + \frac{r(2r-1)}{(r-1)z^2}, \tag{1}$$

with critical cycle

$$0 \longmapsto \infty \longmapsto 1 \longmapsto r \longmapsto 0.$$

Its other critical value is

$$\nu(r) = -\frac{(r^2 - 3r + 1)^2}{4r(r-1)(2r-1)}. \tag{2}$$

The allowed parameter space is

$$\mathcal{P} = \left\{ r \in \mathbb{C} : r(r-1)(2r-1)(r^2 - 3r + 1)(r^2 + r - 1)(3r^2 - 3r + 1) \neq 0 \right\}. \tag{3}$$

In particular, $\mathcal{P}$ is path connected.

Proof. A quadratic rational map with 0 critical, $F(0) = \infty$, and $F(\infty) = 1$ has the form

$$F(z) = 1 + \frac{b}{z} + \frac{c}{z^2}, \qquad c \neq 0.$$

The equations $F(1) = r$ and $F(r) = 0$ give

$$1 + b + c = r, \qquad r^2 + br + c = 0,$$

and solving them gives (1). The second critical value is $1 - b^2/(4c)$, which simplifies to (2). Writing $d(r) = 4r(r-1)(2r-1)$, one also obtains

$$\nu(r) - 1 = -\frac{(r^2 + r - 1)^2}{d(r)}, \qquad \nu(r) - r = -\frac{(3r^2 - 3r + 1)^2}{d(r)}.$$

These formulas give precisely (3). Its complement in $\mathbb{C}$ is finite, so $\mathcal{P}$ is path connected. □

Set

$$R = 100, \qquad v_* = \nu(R) = -\frac{9701^2}{4 \cdot 100 \cdot 99 \cdot 199}.$$

Notice that

$$-12 < v_* < -11. \tag{4}$$

Write

$$A_* = \{0, \infty, 1, R\}, \qquad B_* = A_* \cup \{v_*\}.$$

For a simple closed curve a, let T_a denote the positive Dehn twist about a.

Construction 3.2 (The model curve)**.** On $(\widehat{\mathbb{C}}, B_*)$, take the circle

$$C = \{z : |z - 50| = 55\}.$$

Let α be the embedded polygonal arc

$$[v_*, v_* - 200i] \ \cup \ [v_* - 200i, R - 200i] \ \cup \ [R - 200i, R].$$

Let δ be the boundary of a sufficiently thin regular neighborhood of α, for example its 1/10-neighborhood with corners rounded. Define

$$\boxed{\gamma_* = T_\delta(C).} \tag{5}$$

Here any representative of T_δ supported in a thin collar of δ may be used; the resulting isotopy class of γ_* relative to B_* is independent of that choice.

Thus δ surrounds precisely v_* and R, whereas C, and hence γ_*, separates $\{v_*, \infty\}$ from $\{0, 1, R\}$. The twist in (5) is important: the partition of the marked points alone does not specify the required curve.

Here is the corresponding explicit prescription for the map in the question.

Construction 3.3 (Transport to an arbitrary map)**.** Let m be the normalization in Lemma 3.1, so that $mfm^{-1} = F_{r_0}$ for some $r_0 \in \mathcal{P}$.

Choose a piecewise smooth path $r(t)$ in $\mathcal{P}$ from r_0 to R. This choice can be made entirely explicit: take the polygonal path

$$r_0 \longrightarrow N + iN^2 \longrightarrow R,$$

where N is the least positive integer for which both segments avoid $\mathbb{C} \setminus \mathcal{P}$. Such an N exists, because the forbidden intermediate points lie on finitely many real lines, each meeting the parabola $\{x + ix^2 : x \in \mathbb{R}\}$ finitely often.

Choose an ambient isotopy h_t of $\widehat{\mathbb{C}}$, starting at the identity, such that

$$h_t(0) = 0, \quad h_t(1) = 1, \quad h_t(\infty) = \infty, \quad h_t(r_0) = r(t), \quad h_t(\nu(r_0)) = \nu(r(t)).$$

Define

$$\boxed{\gamma_f = m^{-1}\big(h_1^{-1}(\gamma_*)\big).} \tag{6}$$

Existence of h_t, and independence of its choice up to the relevant isotopy, are justified in Section 7.

Theorem 3.4. *The curve γ_f of Construction 3.3 satisfies*

$$T_{\gamma_f}(\mathcal{D}_f) = \mathcal{D}_f.$$

If $\mathcal{C}_f$ is the connected component containing the standard complex structure of the given rational map, then

$$T_{\gamma_f}^n(\mathcal{C}_f), \qquad n \in \mathbb{Z},$$

are pairwise distinct connected components of $\mathcal{D}_f$. Consequently, the power requested in the question can be taken to be 1.

The Hurwitz-space coordinates used below are those of Hironaka–Koch [4, Proposition 3.1]; we verify the required properties explicitly. Related pairs of equalizing curves occur in Firsova–Kahn–Selinger [2, Section 9, case (12b), arXiv version]. The argument below identifies a particular single twist and proves its infinite component orbit.

4 A covering-space model for the deformation space

Put

$$M = \mathbb{C} \setminus \{0, 1\}, \qquad Y = \{(y, z) \in M^2 : y \neq z\}.$$

These are the moduli spaces of the configurations $(0, \infty, 1, x)$ and $(0, \infty, 1, y, z)$, respectively.

We use the standard facts that the Teichmüller spaces of these labelled punctured spheres are their universal covers, with deck groups their pure mapping class groups. Contractibility and the properly discontinuous action are standard Teichmüller theory; the labelled genus-zero action is free because a Möbius map fixing three labelled points is the identity. See [1, Chapters 10 and 12], or the formulation for marked spheres in [4, Section 2.2].

We let a mapping class h act on markings by

$$h \cdot [\varphi] = [\varphi \circ h^{-1}].$$

Accordingly, a configuration loop extended to an ambient isotopy H_t determines the deck element $[H_1^{-1}]$. Peripheral loops will always be oriented to represent the indicated Dehn twist.

4.1 The explicit Hurwitz covering

For $(x, y) \in \mathbb{C}^2$, define

$$\begin{aligned} D(x, y) &= x(x-1)(x+y-1), \\ Q_0(x, y) &= (x-1)^2 - y, \\ Q_1(x, y) &= x^2 + y - 1, \\ Q_y(x, y) &= (x-1)^2 + (2x-1)y, \end{aligned}$$

and

$$Z(x, y) = -\frac{Q_0(x, y)^2}{4D(x, y)}. \tag{7}$$

Let

$$W = \{(x, y) : y(y-1)D(x, y)Q_0(x, y)Q_1(x, y)Q_y(x, y) \neq 0\}. \tag{8}$$

Lemma 4.1. *The map*

$$q : W \longrightarrow Y, \qquad q(x, y) = (y, Z(x, y)),$$

is a connected, unbranched covering of degree four. The two maps from W to M recording the source configuration and the target configuration with the extra critical value forgotten are

$$p_x(x, y) = x, \qquad p_y(x, y) = y.$$

Proof. A normalized quadratic map satisfying

$$0 \longmapsto \infty, \qquad \infty \longmapsto 1, \qquad 1 \longmapsto y, \qquad x \longmapsto 0$$

must be

$$G_{x,y}(t) = \frac{(t - x)(t - s)}{t^2}, \qquad s = \frac{x + y - 1}{x - 1}. \tag{9}$$

Indeed, the first two conditions make the numerator monic, and $G_{x,y}(1) = y$ determines s. Its second critical value is

$$1 - \frac{(x + s)^2}{4xs} = Z(x, y).$$

Direct expansion gives

$$Z - 1 = -\frac{Q_1^2}{4D}, \qquad Z - y = -\frac{Q_y^2}{4D}, \qquad \frac{\partial Z}{\partial x} = -\frac{Q_0 Q_1 Q_y}{4D^2}. \tag{10}$$

Thus (8) says exactly that the source and target markings are distinct and that the second critical value avoids the other target markings.

For fixed $(y, z) \in Y$, the equation $Z(x, y) = z$ is

$$Q_0(x, y)^2 + 4zD(x, y) = 0,$$

a monic polynomial equation of degree four in x. None of its roots is 0, 1, or $1 - y$: at these points Q_0 is nonzero because $y \neq 0, 1$. The identities (10) show that all four roots are simple when $z \notin \{0, 1, y\}$. The implicit function theorem gives four local inverse branches of q. Hence q is an unbranched four-sheeted covering.

> **Reviewer 1**
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> 'Hence' skips a step. Local inverse branches over every point, together with a constant fibre cardinality of four, are what make q a covering (intersect the images of disjoint neighbourhoods of the four preimages). Also say that $q(W) \subset Y$.

Finally, W is path connected. Given two of its points, restrict the nonvanishing polynomial defining W to the complex affine line through them. It is not identically zero, and therefore excludes only finitely many points of that line. The two points can be joined in the complement of those finitely many points.

The descriptions of p_x and p_y follow from (9). □

Lemma 4.2. *For each $r \in \mathcal{P}$, there is a universal covering*

$$U_r : \mathrm{Teich}(B_r) \longrightarrow W, \qquad B_r = \{0, \infty, 1, r, \nu(r)\},$$

taking the standard marked structure τ_r to (r, r), such that

$$qU_r = \pi_{B_r}, \qquad \pi_{A_r} F_r^* = p_x U_r, \qquad \pi_{A_r} i_* = p_y U_r, \tag{11}$$

where $A_r = \{0, \infty, 1, r\}$.

Proof. Given a marked complex structure $[\varphi] \in \mathrm{Teich}(B_r)$, normalize φ to fix $0, \infty, 1$. Uniformize the pullback complex structure, obtaining a normalized marking ψ for which

$$G = \varphi \circ F_r \circ \psi^{-1}$$

is rational. This is the usual uniformization description of the pullback map; see [4, Sections 2.1–2.2, Proposition 2.1]. Set

$$x = \psi(r), \qquad y = \varphi(r), \qquad z = \varphi(\nu(r)).$$

The map G has precisely the normalization in (9), so $z = Z(x, y)$. This gives the continuous lift U_r and all identities in (11).

Since $\mathrm{Teich}(B_r)$ is the universal cover of Y and q is a connected covering, its lift U_r is the universal cover of W. This is the standard composition and lifting property of coverings; see [3, Section 1.3]. □

4.2 The stabilizer of one component

For the rest of the model argument, use $r = R$, write $U = U_R$, and take the basepoints

$$\tau_* = \tau_R, \qquad w_* = (R, R).$$

The injection q_* identifies

$$L := \pi_1(W, w_*)$$

with a subgroup of the deck group of $\mathrm{Teich}(B_*) \to Y$.

Define

$$d : \mathcal{P} \longrightarrow W, \qquad d(r) = (r, r), \qquad K := \mathrm{im}\big(d_* : \pi_1(\mathcal{P}, R) \to L\big).$$

The formulas defining W show that its diagonal is exactly $d(\mathcal{P})$.

Lemma 4.3. *Let $\mathcal{C}_*$ be the component of $\mathcal{D}_{F_R}$ containing τ_*. Then:*

1. *$\mathcal{D}_{F_R}$ is a union of connected components of $U^{-1}(d(\mathcal{P}))$, and each such component is path connected;*
2. *for $g \in L$,*
$$g(\mathcal{C}_*) = \mathcal{C}_* \quad \Longleftrightarrow \quad g \in K;$$
3. *if*
$$(p_x)_*(g) = (p_y)_*(g) \quad \text{in } \pi_1(M, R),$$
then $g(\mathcal{D}_{F_R}) = \mathcal{D}_{F_R}$.

Proof. On $U^{-1}(d(\mathcal{P}))$, the maps F_R^* and i_* are lifts to $\mathrm{Teich}(A_*)$ of the same map to M, by (11). On a connected component, two lifts either agree everywhere or nowhere: if they agree at one point, uniqueness of lifting makes them agree throughout that component. Since $U^{-1}(d(\mathcal{P})) \to d(\mathcal{P})$ is a covering of a Riemann surface, its components are path connected. This proves the first assertion.

A deck element $g \in L$ carries τ_* into $\mathcal{C}_*$ exactly when there is a path in $U^{-1}(d(\mathcal{P}))$ from τ_* to $g(\tau_*)$. Projecting this path gives a loop in $d(\mathcal{P})$ representing g; conversely, such a loop lifts to such a path. This proves the second assertion.

For the last assertion, lifting uniqueness gives the equivariance relations

$$F_R^* \circ g = (p_x)_*(g) \circ F_R^*, \qquad i_* \circ g = (p_y)_*(g) \circ i_*,$$

where the elements on the right act as deck transformations of $\mathrm{Teich}(A_*)$. Equality of the two deck elements implies $g(\mathcal{D}_{F_R}) \subseteq \mathcal{D}_{F_R}$; applying the same argument to g^{-1} gives equality. □

This is the covering-space form of the special-liftable and component-stabilizer descriptions in [4, Sections 2.8–2.9]. The covering facts used in its proof are the lifting and deck-transformation results in [3, Section 1.3, especially Proposition 1.39].

5 A free-group quotient detecting movement of components

Enlarge W to the five-line complement

$$X = \{(x, y) \in M^2 : x + y \neq 1\},$$

and let $j : W \hookrightarrow X$ be inclusion. Put

$$\mathcal{P}_0 = \mathbb{C} \setminus \{0, 1, \tfrac{1}{2}\}, \qquad e : \mathcal{P}_0 \longrightarrow X, \quad e(r) = (r, r).$$

This auxiliary complement also appears in [4, Section 2.12]. The properties needed here have short direct proofs.

Lemma 5.1. *The map e_* is injective. Consequently*

$$H := e_*\pi_1(\mathcal{P}_0, R)$$

is a free group of rank three. Moreover,

$$j_*(K) = H. \tag{12}$$

Proof. Consider

$$\Phi : X \longrightarrow M, \qquad \Phi(x, y) = \frac{xy}{x + y - 1}.$$

It avoids 0 because $xy \neq 0$, and it avoids 1 because

$$xy - (x + y - 1) = (x - 1)(y - 1) \neq 0.$$

On the diagonal,

$$\Phi e(r) = \psi(r) := \frac{r^2}{2r - 1}.$$

The map $\psi : \mathcal{P}_0 \to M$ is an unbranched double covering. Indeed, for $w \notin \{0, 1\}$ its inverse points are the two distinct roots of

$$r^2 - 2wr + w = 0,$$

and neither root is 0, 1, or 1/2. Thus ψ_* is injective, as is the fundamental-group homomorphism of any covering. Since $\Phi_* e_* = \psi_*$, e_* is injective.

The plane with three points removed has fundamental group free of rank three: disjoint small circles about the punctures, joined to a basepoint by a tree of arcs, give the usual graph model. Thus $H \cong F_3$.

Finally, $\mathcal{P}$ is obtained from $\mathcal{P}_0$ by deleting finitely many additional points. The inclusion $\mathcal{P} \hookrightarrow \mathcal{P}_0$ induces a surjection on fundamental groups: a loop can be perturbed in small disks to avoid those additional points. Using $jd = e|_{\mathcal{P}}$ proves (12). □

We will also use the winding-number homomorphism

$$\lambda : \pi_1(X, w_*) \longrightarrow \mathbb{Z}, \qquad \lambda([\beta]) = \operatorname{wind}_0\big((x + y - 1) \circ \beta\big). \tag{13}$$

Lemma 5.2 (A free-group centralizer criterion)**.** *Let F be a free group and $\theta : F \to G$ a homomorphism. If $\theta(t)$ has infinite order, then*

$$\{a \in F : at = ta,\ \theta(a) = 1\} = \{1\}.$$

Proof. The centralizer of a nonidentity element of a free group is infinite cyclic. To recall the argument, let the free group act on its Cayley tree. Writing an element as a conjugate of a cyclically reduced word exhibits its translation axis, which is unique. A commuting element preserves this axis and acts on it by a translation, not a reflection. Signed translation distance embeds the centralizer into $\mathbb{Z}$: an element with zero translation fixes a vertex, and the left action of a free group on its Cayley tree is free on vertices. The centralizer is therefore cyclic.

Write this centralizer as $\langle b \rangle$, with $t = b^k$, $k \neq 0$. Since $\theta(t)$ has infinite order, so does $\theta(b)$. Hence θ is injective on $\langle b \rangle$, proving the claim. □

6 Verification for the explicit model curve

6.1 Elementary twist conventions

We will use three elementary facts:

$$T_{h(a)} = hT_a h^{-1},$$

twists about disjoint curves commute, and a twist about the boundary of a disk containing exactly two marked points is represented by moving one of those points once around the other.

For completeness, the first two facts follow by transporting, or choosing disjoint, annular supports. For the third, identify the disk with a round disk, place the fixed puncture at its center, and use

$$z \longmapsto e^{2\pi i t \chi(|z|)} z,$$

where χ is one on a disk containing the moving puncture and zero near the boundary. The puncture travels once around the center, and the endpoint is one boundary Dehn twist, with sign determined by the chosen conventions. Reversing the loop gives the other sign. These are also the standard twist and point-pushing facts in [1, Sections 3.3 and 4.2].

Let

$$\eta = \{z : |z| = 2\}, \qquad u = T_\delta, \qquad g = T_{\gamma_*} = uT_C u^{-1}.$$

The disk bounded by η lies strictly inside the disk bounded by C, and the neighborhood defining δ is disjoint from that disk. Therefore

$$\gamma_* \cap \eta = \varnothing \tag{14}$$

for suitable representatives.

6.2 The monodromy of the required twist

Lemma 6.1. *The element $g = T_{\gamma_*}$ belongs to L and satisfies*

$$(p_x)_*(g) = 1, \qquad (p_y)_*(g) = 1, \qquad |\lambda(j_* g)| = 1. \tag{15}$$

In particular,

$$g(\mathcal{D}_{F_R}) = \mathcal{D}_{F_R}.$$

Proof. Keep $y = R$ fixed and consider the rational map

$$x \longmapsto Z(x, R).$$

By (10), its branch values are $0, 1, R$. Each has two simple ramification points. Its four poles

$$0, \quad 1, \quad 1 - R, \quad \infty$$

are simple.

The Möbius involution

$$J(x) = \frac{(1 - R)(x - 1)}{x + R - 1} \tag{16}$$

satisfies

$$J^2 = \mathrm{id}, \qquad Z(J(x), R) = Z(x, R). \tag{17}$$

Both identities follow by substitution. Its fixed-point equation is

$$x^2 + 2(R - 1)x + 1 - R = 0,$$

which is exactly $Q_y(x, R) = 0$. Thus its fixed points are the two ramification points over R.

It follows that the sheet monodromy of a based meridian about the critical value R is J. Indeed, close to either of these ramification points a meridian interchanges the two local inverse branches, as does J.

> **Reviewer 1**
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> The real reason the monodromy is J for any access path is that J is a deck transformation of $x \mapsto Z(x, R)$, so it commutes with path lifting; locally it swaps the two branches at each of its fixed points. Saying this makes 'Transport to the basepoint commutes with J' a proof rather than an assertion.

Transport to the basepoint commutes with J by (17). The assertion holds for either orientation of the meridian.

By the point-pushing description, $u = T_\delta$ is represented by such a meridian about $z = R$, with $y = R$ fixed. Consequently its lift to the four-sheeted covering starts at $x = R$ and ends at $x = J(R)$.

The twist T_C is represented by a meridian about $z = \infty$. Its access path from v_* to infinity may be chosen along the negative real axis, since that ray lies in the complementary disk of C containing v_* and ∞.

Starting on the sheet $x = R$, the lift of this access path tends to $x = \infty$. To see this directly, for real $x \geq R$ all three quantities $Q_0(x, R), Q_1(x, R), Q_y(x, R)$ are positive. Equation (10) therefore gives

$$\frac{d}{dx} Z(x, R) < 0.$$

Moreover,

$$Z(R, R) = v_*, \qquad Z(x, R) \longrightarrow -\infty \quad (x \longrightarrow +\infty).$$

Thus this real inverse branch parametrizes precisely the required lift.

Starting instead at $J(R)$, the lifted access path is its image under J, and therefore tends to

$$J(\infty) = 1 - R.$$

Because this is a simple pole, the infinity meridian lifts to a single meridian about $x = 1 - R$, and returns to the same sheet.

Now represent

$$g = u T_C u^{-1}$$

> **Reviewer 1**
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> Whether the conjugated configuration loop represents uT_Cu^{-1} or $u^{-1}T_Cu$ depends on conventions (deck elements, concatenation) that are not tracked. This is harmless: J is an involution and the argument is symmetric, so both $T_\delta^{\pm1}(C)$ work. But it should be said, since the theorem names $T_\delta(C)$ specifically.

by the corresponding conjugated configuration loop. Its lift first changes the sheet from R to $J(R)$, then makes the just-described meridian about $x = 1-R$, and finally returns along the inverse conjugating path. Hence the lift is closed, proving $g \in L$. In the fiber $y = R$, its homotopy class is a based conjugate of a single meridian about $x = 1 - R$.

Since $1 - R \notin \{0, 1\}$, that meridian is null-homotopic after projection to the x-coordinate space M. Its y-coordinate is constant. On the other hand, $x + y - 1 = x + R - 1$ winds once, with one of the two signs, about zero. This proves (15).

The final assertion follows from Lemma 4.3(3). □

6.3 A commuting twist that does preserve the base component

Lemma 6.2. *The diagonal loop*

$$t(s) = \big(Re^{2\pi is}, Re^{2\pi is}\big), \qquad 0 \le s \le 1, \tag{18}$$

defines an element $t \in K$ whose mapping class is T_η or T_η^{-1}. Moreover, $(p_y)_*(j_*t)$ has infinite order in $\pi_1(M, R)$.

Proof. The defining factors of (3) do not vanish on $|r| = R = 100$, by the triangle inequality. Thus (18) is a loop in $d(\mathcal{P})$, so $t \in K$.

We identify its mapping class. A calculation from (2) gives

$$\frac{\nu(r)}{r} + \frac{1}{8} = \frac{9r^3 - 21r^2 + 12r - 2}{8r^2(r-1)(2r-1)}.$$

Consequently, when $|r| = R = 100$,

$$\left|\frac{\nu(r)}{r} + \frac{1}{8}\right| \le \frac{9R^3 + 21R^2 + 12R + 2}{8R^2(R-1)(2R-1)} < \frac{1}{100}. \tag{19}$$

Set $a_0 = \nu(R)/R$. The projected configuration loop

$$\big(r(s), \nu(r(s))\big), \qquad r(s) = Re^{2\pi is},$$

has the based homotopy

$$\left(r(s),\ r(s)\left((1-a)\frac{\nu(r(s))}{r(s)} + aa_0\right)\right), \qquad 0 \le a \le 1.$$

Indeed, the expression in parentheses stays in the convex disk of radius $1/100$ about $-1/8$. The moving critical value therefore has modulus greater than 11.5, and is never 0, 1, or $r(s)$. At the end of the homotopy the loop is

$$\big(Re^{2\pi is}, v_*e^{2\pi is}\big).$$

Choose a smooth radial function χ equal to zero on $[0, 2]$ and equal to one on $[10, \infty)$. The ambient isotopy

$$H_s(z) = e^{2\pi is\chi(|z|)}z, \qquad H_s(\infty) = \infty,$$

realizes this last configuration loop. Its endpoint is one Dehn twist in the unmarked annulus $2 < |z| < 10$, and hence is a twist about η. Passing to the deck-action convention may invert it, which proves the asserted description of t.

Finally, the y-coordinate of (18) winds once about zero. Its image in $\pi_1(M, R)$ therefore has infinite order. □

6.4 Infinitely many different components

Proposition 6.3. *The components*

$$g^n(\mathcal{C}_*), \qquad n \in \mathbb{Z},$$

are pairwise distinct.

Proof. By Lemma 6.1, g preserves the deformation space. Suppose that

$$g^n(\mathcal{C}_*) = \mathcal{C}_*$$

for some nonzero integer n. Lemma 4.3(2) gives $g^n \in K$, and hence, by Lemma 5.1,

$$j_*(g^n) \in H.$$

Also $j_*t \in H$.

The curves γ_* and η are disjoint by (14). Thus g commutes with the mapping class of t. Since q_* is injective, they commute in L, and their images commute in $\pi_1(X, w_*)$.

Apply Lemma 5.2 to the free group H and the homomorphism

$$(p_y)_*|_H : H \longrightarrow \pi_1(M, R).$$

By Lemma 6.2, the image of j_*t has infinite order. By Lemma 6.1, the image of $j_*(g^n)$ is the identity. The centralizer criterion therefore gives

$$j_*(g^n) = 1.$$

But (15) and (13) imply

$$\big|\lambda\big(j_*(g^n)\big)\big| = |n| > 0,$$

a contradiction.

Thus no nonzero power of g stabilizes $\mathcal{C}_*$. If two of the displayed components were equal, their index difference would give such a power. They are therefore pairwise distinct. □

7 Transport and completion of the proof

We first justify the isotopy required in Construction 3.3. Along the compact parameter path, the five labelled points

$$0, \quad 1, \quad \infty, \quad r(t), \quad \nu(r(t))$$

remain distinct. On each smooth portion of the path, choose disjoint small disks around the two moving finite points, avoiding 0 and 1. A compactly supported time-dependent vector field which equals the velocity of each moving point near that point has a flow tracking both points and fixing $0, 1, \infty$. Concatenating these flows constructs h_t.

If h_t and $\widetilde{h}_t$ are two such extensions of the same configuration path, then

$$\widetilde{h}_t^{-1} h_t$$

fixes the initial labelled configuration pointwise for every t. Thus the two endpoint pullbacks of γ_* are isotopic relative to that configuration. This proves that the construction specifies an isotopy class without any further choice of ambient isotopy.

Lemma 7.1. *For the isotopy h_t of Construction 3.3, put $h = h_1$. There is a homeomorphism*

$$\Theta : \mathrm{Teich}(B_*) \longrightarrow \mathrm{Teich}(B_{r_0}), \qquad \Theta([\chi]) = [\chi \circ h],$$

such that

$$\Theta(\mathcal{D}_{F_R}) = \mathcal{D}_{F_{r_0}}.$$

It takes $\mathcal{C}_*$ onto the component containing the standard complex structure for F_{r_0}, and intertwines T_{γ_*} with $T_{h^{-1}(\gamma_*)}$.

Proof. Consider the path

$$\tau(t) = [h_t] \in \mathrm{Teich}(B_{r_0}).$$

Its projection to Y is

$$\big(r(t), \nu(r(t))\big).$$

The diagonal path $(r(t), r(t))$ is a lift of this path to W. By Lemma 4.2 and uniqueness of lifting,

$$U_{r_0}(\tau(t)) = (r(t), r(t)).$$

The two paths $F_{r_0}^*(\tau(t))$ and $i_*(\tau(t))$ are therefore lifts to $\mathrm{Teich}(A_{r_0})$ of the same path $r(t)$ in M. They start at the same point, so they coincide for every t. Consequently

$$\tau(t) \in \mathcal{D}_{F_{r_0}}.$$

At $t = 1$, normalized uniformization of the pullback gives a marking h' such that

$$F_R \circ h' = h \circ F_{r_0}. \tag{20}$$

The rational map on the left is F_R because its Hurwitz coordinates are (R, R), and (9) uniquely determines that normalized map.

> **Reviewer 1**
>
> This again uses $G_{R,R} = F_R$. Also, $h' \simeq h$ rel A_{r_0} uses that both are normalized to fix $0, 1, \infty$, so the Möbius ambiguity is trivial. Worth one clause each.

Since $\tau(1)$ lies in the deformation space,

$$h' \simeq h \quad \text{relative to } A_{r_0}.$$

Let

$$\Theta_A : \mathrm{Teich}(A_*) \longrightarrow \mathrm{Teich}(A_{r_0})$$

be change of marking by $h|_{A_{r_0}}$. Naturality of pullback applied to (20), and the relative isotopy of h' and h, give

$$F_{r_0}^* \Theta = \Theta_A F_R^*, \qquad i_* \Theta = \Theta_A i_*.$$

Since Θ_A is a homeomorphism, these identities imply

$$\Theta(\mathcal{D}_{F_R}) = \mathcal{D}_{F_{r_0}}.$$

Furthermore, $\Theta(\tau_*) = [h] = \tau(1)$ lies in the same component as the standard structure $\tau(0)$. Hence Θ carries $\mathcal{C}_*$ onto that component.

Finally, change of marking conjugates a mapping class a to $h^{-1}ah$. Since conjugating a Dehn twist gives the Dehn twist about the inverse-image curve, Θ intertwines the stated twists. □

Proof of Theorem 3.4. Lemma 6.1 and Proposition 6.3 prove the theorem for F_R and γ_*. Lemma 7.1 transfers both preservation of the deformation space and pairwise distinctness of the component orbit to F_{r_0} and $h^{-1}(\gamma_*)$. Undoing the Möbius normalization transfers these conclusions to f and the curve (6). □

Remark 7.2. The proof distinguishes two assertions which should not be conflated. The vanishing of both coordinate monodromies proves that the twist preserves the *whole* deformation space. Its failure to preserve any nonzero-power iterate of the base component is a separate fact, detected by the winding homomorphism (13) together with a cyclic-centralizer argument in the free group H.

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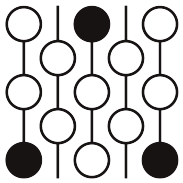
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Review of Problem 7 – Astra

First Proof Third Batch
Reviewer 2
October 8–9, 2026



1 Review

The article produces the desired explicit description of a simple closed curve γ, such that the Dehn twist around γ has an infinite orbit on the components of the deformation space of quadratic maps with a critical 4-cycle.

Correctness. I think the solution is correct.
Score: 1

Exposition. The argument is written in a concise and reasonable manner. The proposed solution is less computationally involved than the others, making it more readable.
Score: 0

Attributions. The attributions are adequate.
Score: 0

2 Submission with inline comments

Abstract

We describe a simple closed curve for which the Dehn twist itself, not merely a proper power, preserves the deformation space in the question and has an infinite orbit on its connected components. The curve is first specified for one explicit normalized rational map and is then transported to an arbitrary map in the family. The proof uses an explicit Hurwitz covering and a free-group quotient of the stabilizer of a component. It does not assume that the components of the deformation space are simply connected.

3 The explicit curve and the statement

All marked points are labelled. Thus mapping class groups below are pure mapping class groups, and moduli spaces are moduli spaces of ordered configurations.

Lemma 3.1 (Normal form)**.** *After the unique Möbius normalization*

$$c_1 = 0, \qquad f(c_1) = \infty, \qquad f^{\circ 2}(c_1) = 1,$$

the map in the question is

$$F_r(z) = 1 - \frac{r^2 + r - 1}{(r-1)z} + \frac{r(2r-1)}{(r-1)z^2}, \tag{1}$$

with critical cycle

$$0 \longmapsto \infty \longmapsto 1 \longmapsto r \longmapsto 0.$$

Its other critical value is

$$\nu(r) = -\frac{(r^2 - 3r + 1)^2}{4r(r-1)(2r-1)}. \tag{2}$$

The allowed parameter space is

$$\mathcal{P} = \left\{ r \in \mathbb{C} : r(r-1)(2r-1)(r^2 - 3r + 1)(r^2 + r - 1)(3r^2 - 3r + 1) \neq 0 \right\}. \tag{3}$$

In particular, $\mathcal{P}$ is path connected.

Proof. A quadratic rational map with 0 critical, $F(0) = \infty$, and $F(\infty) = 1$ has the form

$$F(z) = 1 + \frac{b}{z} + \frac{c}{z^2}, \qquad c \neq 0.$$

The equations $F(1) = r$ and $F(r) = 0$ give

$$1 + b + c = r, \qquad r^2 + br + c = 0,$$

and solving them gives (1). The second critical value is $1 - b^2/(4c)$, which simplifies to (2). Writing $d(r) = 4r(r-1)(2r-1)$, one also obtains

$$\nu(r) - 1 = -\frac{(r^2 + r - 1)^2}{d(r)}, \qquad \nu(r) - r = -\frac{(3r^2 - 3r + 1)^2}{d(r)}.$$

These formulas give precisely (3). Its complement in $\mathbb{C}$ is finite, so $\mathcal{P}$ is path connected. □

Set

$$R = 100, \qquad v_* = \nu(R) = -\frac{9701^2}{4 \cdot 100 \cdot 99 \cdot 199}.$$

Notice that

$$-12 < v_* < -11. \tag{4}$$

Write

$$A_* = \{0, \infty, 1, R\}, \qquad B_* = A_* \cup \{v_*\}.$$

For a simple closed curve a, let T_a denote the positive Dehn twist about a.

Construction 3.2 (The model curve)**.** On $(\widehat{\mathbb{C}}, B_*)$, take the circle

$$C = \{z : |z - 50| = 55\}.$$

Let α be the embedded polygonal arc

$$[v_*, v_* - 200i] \ \cup \ [v_* - 200i, R - 200i] \ \cup \ [R - 200i, R].$$

Let δ be the boundary of a sufficiently thin regular neighborhood of α, for example its 1/10-neighborhood with corners rounded. Define

$$\boxed{\gamma_* = T_\delta(C).} \tag{5}$$

Here any representative of T_δ supported in a thin collar of δ may be used; the resulting isotopy class of γ_* relative to B_* is independent of that choice.

Thus δ surrounds precisely v_* and R, whereas C, and hence γ_*, separates $\{v_*, \infty\}$ from $\{0, 1, R\}$. The twist in (5) is important: the partition of the marked points alone does not specify the required curve.

Here is the corresponding explicit prescription for the map in the question.

Construction 3.3 (Transport to an arbitrary map)**.** Let m be the normalization in Lemma 3.1, so that $mfm^{-1} = F_{r_0}$ for some $r_0 \in \mathcal{P}$.

Choose a piecewise smooth path $r(t)$ in $\mathcal{P}$ from r_0 to R. This choice can be made entirely explicit: take the polygonal path

$$r_0 \longrightarrow N + iN^2 \longrightarrow R,$$

where N is the least positive integer for which both segments avoid $\mathbb{C} \setminus \mathcal{P}$. Such an N exists, because the forbidden intermediate points lie on finitely many real lines, each meeting the parabola $\{x + ix^2 : x \in \mathbb{R}\}$ finitely often.

Choose an ambient isotopy h_t of $\widehat{\mathbb{C}}$, starting at the identity, such that

$$h_t(0) = 0, \quad h_t(1) = 1, \quad h_t(\infty) = \infty, \quad h_t(r_0) = r(t), \quad h_t(\nu(r_0)) = \nu(r(t)).$$

Define

$$\boxed{\gamma_f = m^{-1}\big(h_1^{-1}(\gamma_*)\big).} \tag{6}$$

Existence of h_t, and independence of its choice up to the relevant isotopy, are justified in Section 7.

Theorem 3.4. *The curve γ_f of Construction 3.3 satisfies*

$$T_{\gamma_f}(\mathcal{D}_f) = \mathcal{D}_f.$$

If $\mathcal{C}_f$ is the connected component containing the standard complex structure of the given rational map, then

$$T_{\gamma_f}^n(\mathcal{C}_f), \qquad n \in \mathbb{Z},$$

are pairwise distinct connected components of $\mathcal{D}_f$. Consequently, the power requested in the question can be taken to be 1.

The Hurwitz-space coordinates used below are those of Hironaka–Koch [4, Proposition 3.1]; we verify the required properties explicitly. Related pairs of equalizing curves occur in Firsova–Kahn–Selinger [2, Section 9, case (12b), arXiv version]. The argument below identifies a particular single twist and proves its infinite component orbit.

4 A covering-space model for the deformation space

Put

$$M = \mathbb{C} \setminus \{0, 1\}, \qquad Y = \{(y, z) \in M^2 : y \neq z\}.$$

These are the moduli spaces of the configurations $(0, \infty, 1, x)$ and $(0, \infty, 1, y, z)$, respectively.

We use the standard facts that the Teichmüller spaces of these labelled punctured spheres are their universal covers, with deck groups their pure mapping class groups. Contractibility and the properly discontinuous action are standard Teichmüller theory; the labelled genus-zero action is free because a Möbius map fixing three labelled points is the identity. See [1, Chapters 10 and 12], or the formulation for marked spheres in [4, Section 2.2].

We let a mapping class h act on markings by

$$h \cdot [\varphi] = [\varphi \circ h^{-1}].$$

Accordingly, a configuration loop extended to an ambient isotopy H_t determines the deck element $[H_1^{-1}]$. Peripheral loops will always be oriented to represent the indicated Dehn twist.

4.1 The explicit Hurwitz covering

For $(x, y) \in \mathbb{C}^2$, define

$$\begin{aligned} D(x, y) &= x(x-1)(x+y-1), \\ Q_0(x, y) &= (x-1)^2 - y, \\ Q_1(x, y) &= x^2 + y - 1, \\ Q_y(x, y) &= (x-1)^2 + (2x-1)y, \end{aligned}$$

and

$$Z(x, y) = -\frac{Q_0(x, y)^2}{4D(x, y)}. \tag{7}$$

Let

$$W = \{(x, y) : y(y-1)D(x, y)Q_0(x, y)Q_1(x, y)Q_y(x, y) \neq 0\}. \tag{8}$$

Lemma 4.1. *The map*

$$q : W \longrightarrow Y, \qquad q(x, y) = (y, Z(x, y)),$$

is a connected, unbranched covering of degree four. The two maps from W to M recording the source configuration and the target configuration with the extra critical value forgotten are

$$p_x(x, y) = x, \qquad p_y(x, y) = y.$$

Proof. A normalized quadratic map satisfying

$$0 \longmapsto \infty, \qquad \infty \longmapsto 1, \qquad 1 \longmapsto y, \qquad x \longmapsto 0$$

must be

$$G_{x,y}(t) = \frac{(t - x)(t - s)}{t^2}, \qquad s = \frac{x + y - 1}{x - 1}. \tag{9}$$

Indeed, the first two conditions make the numerator monic, and $G_{x,y}(1) = y$ determines s. Its second critical value is

$$1 - \frac{(x + s)^2}{4xs} = Z(x, y).$$

Direct expansion gives

$$Z - 1 = -\frac{Q_1^2}{4D}, \qquad Z - y = -\frac{Q_y^2}{4D}, \qquad \frac{\partial Z}{\partial x} = -\frac{Q_0 Q_1 Q_y}{4D^2}. \tag{10}$$

Thus (8) says exactly that the source and target markings are distinct and that the second critical value avoids the other target markings.

For fixed $(y, z) \in Y$, the equation $Z(x, y) = z$ is

$$Q_0(x, y)^2 + 4zD(x, y) = 0,$$

a monic polynomial equation of degree four in x. None of its roots is 0, 1, or $1 - y$: at these points Q_0 is nonzero because $y \neq 0, 1$. The identities (10) show that all four roots are simple when $z \notin \{0, 1, y\}$. The implicit function theorem gives four local inverse branches of q. Hence q is an unbranched four-sheeted covering.

Finally, W is path connected. Given two of its points, restrict the nonvanishing polynomial defining W to the complex affine line through them. It is not identically zero, and therefore excludes only finitely many points of that line. The two points can be joined in the complement of those finitely many points.

The descriptions of p_x and p_y follow from (9). □

Lemma 4.2. *For each $r \in \mathcal{P}$, there is a universal covering*

$$U_r : \mathrm{Teich}(B_r) \longrightarrow W, \qquad B_r = \{0, \infty, 1, r, \nu(r)\},$$

taking the standard marked structure τ_r to (r, r), such that

$$qU_r = \pi_{B_r}, \qquad \pi_{A_r} F_r^* = p_x U_r, \qquad \pi_{A_r} i_* = p_y U_r, \tag{11}$$

where $A_r = \{0, \infty, 1, r\}$.

Proof. Given a marked complex structure $[\varphi] \in \mathrm{Teich}(B_r)$, normalize φ to fix $0, \infty, 1$. Uniformize the pullback complex structure, obtaining a normalized marking ψ for which

$$G = \varphi \circ F_r \circ \psi^{-1}$$

is rational. This is the usual uniformization description of the pullback map; see [4, Sections 2.1–2.2, Proposition 2.1]. Set

$$x = \psi(r), \qquad y = \varphi(r), \qquad z = \varphi(\nu(r)).$$

The map G has precisely the normalization in (9), so $z = Z(x, y)$. This gives the continuous lift U_r and all identities in (11).

Since $\mathrm{Teich}(B_r)$ is the universal cover of Y and q is a connected covering, its lift U_r is the universal cover of W. This is the standard composition and lifting property of coverings; see [3, Section 1.3]. □

4.2 The stabilizer of one component

For the rest of the model argument, use $r = R$, write $U = U_R$, and take the basepoints

$$\tau_* = \tau_R, \qquad w_* = (R, R).$$

The injection q_* identifies

$$L := \pi_1(W, w_*)$$

with a subgroup of the deck group of $\mathrm{Teich}(B_*) \to Y$.

Define

$$d : \mathcal{P} \longrightarrow W, \qquad d(r) = (r, r), \qquad K := \mathrm{im}\big(d_* : \pi_1(\mathcal{P}, R) \to L\big).$$

The formulas defining W show that its diagonal is exactly $d(\mathcal{P})$.

Lemma 4.3. *Let $\mathcal{C}_*$ be the component of $\mathcal{D}_{F_R}$ containing τ_*. Then:*

1. *$\mathcal{D}_{F_R}$ is a union of connected components of $U^{-1}(d(\mathcal{P}))$, and each such component is path connected;*
2. *for $g \in L$,*
$$g(\mathcal{C}_*) = \mathcal{C}_* \quad \Longleftrightarrow \quad g \in K;$$
3. *if*
$$(p_x)_*(g) = (p_y)_*(g) \quad \text{in } \pi_1(M, R),$$
then $g(\mathcal{D}_{F_R}) = \mathcal{D}_{F_R}$.

Proof.

> **Reviewer 2**
>
> The first assertion could be explained better

On $U^{-1}(d(\mathcal{P}))$, the maps F_R^* and i_* are lifts to $\mathrm{Teich}(A_*)$ of the same map to M, by (11). On a connected component, two lifts either agree everywhere or nowhere: if they agree at one point, uniqueness of lifting makes them agree throughout that component. Since $U^{-1}(d(\mathcal{P})) \to d(\mathcal{P})$ is a covering of a Riemann surface, its components are path connected. This proves the first assertion.

A deck element $g \in L$ carries τ_* into $\mathcal{C}_*$ exactly when there is a path in $U^{-1}(d(\mathcal{P}))$ from τ_* to $g(\tau_*)$. Projecting this path gives a loop in $d(\mathcal{P})$ representing g; conversely, such a loop lifts to such a path. This proves the second assertion.

For the last assertion, lifting uniqueness gives the equivariance relations

$$F_R^* \circ g = (p_x)_*(g) \circ F_R^*, \qquad i_* \circ g = (p_y)_*(g) \circ i_*,$$

where the elements on the right act as deck transformations of $\mathrm{Teich}(A_*)$. Equality of the two deck elements implies $g(\mathcal{D}_{F_R}) \subseteq \mathcal{D}_{F_R}$; applying the same argument to g^{-1} gives equality. □

This is the covering-space form of the special-liftable and component-stabilizer descriptions in [4, Sections 2.8–2.9]. The covering facts used in its proof are the lifting and deck-transformation results in [3, Section 1.3, especially Proposition 1.39].

5 A free-group quotient detecting movement of components

Enlarge W to the five-line complement

$$X = \{(x, y) \in M^2 : x + y \neq 1\},$$

and let $j : W \hookrightarrow X$ be inclusion. Put

$$\mathcal{P}_0 = \mathbb{C} \setminus \{0, 1, \tfrac{1}{2}\}, \qquad e : \mathcal{P}_0 \longrightarrow X, \quad e(r) = (r, r).$$

This auxiliary complement also appears in [4, Section 2.12]. The properties needed here have short direct proofs.

Lemma 5.1. *The map e_* is injective. Consequently*

$$H := e_*\pi_1(\mathcal{P}_0, R)$$

is a free group of rank three. Moreover,

$$j_*(K) = H. \tag{12}$$

Proof. Consider

$$\Phi : X \longrightarrow M, \qquad \Phi(x, y) = \frac{xy}{x + y - 1}.$$

It avoids 0 because $xy \neq 0$, and it avoids 1 because

$$xy - (x + y - 1) = (x - 1)(y - 1) \neq 0.$$

On the diagonal,

$$\Phi e(r) = \psi(r) := \frac{r^2}{2r - 1}.$$

The map $\psi : \mathcal{P}_0 \to M$ is an unbranched double covering. Indeed, for $w \notin \{0, 1\}$ its inverse points are the two distinct roots of

$$r^2 - 2wr + w = 0,$$

and neither root is 0, 1, or 1/2. Thus ψ_* is injective, as is the fundamental-group homomorphism of any covering. Since $\Phi_* e_* = \psi_*$, e_* is injective.

The plane with three points removed has fundamental group free of rank three: disjoint small circles about the punctures, joined to a basepoint by a tree of arcs, give the usual graph model. Thus $H \cong F_3$.

Finally, $\mathcal{P}$ is obtained from $\mathcal{P}_0$ by deleting finitely many additional points. The inclusion $\mathcal{P} \hookrightarrow \mathcal{P}_0$ induces a surjection on fundamental groups: a loop can be perturbed in small disks to avoid those additional points. Using $jd = e|_{\mathcal{P}}$ proves (12). □

We will also use the winding-number homomorphism

$$\lambda : \pi_1(X, w_*) \longrightarrow \mathbb{Z}, \qquad \lambda([\beta]) = \mathrm{wind}_0\big((x+y-1)\circ\beta\big). \tag{13}$$

Lemma 5.2 (A free-group centralizer criterion)**.** *Let F be a free group and $\theta : F \to G$ a homomorphism. If $\theta(t)$ has infinite order, then*

$$\{a \in F : at = ta,\ \theta(a) = 1\} = \{1\}.$$

Proof. The centralizer of a nonidentity element of a free group is infinite cyclic. To recall the argument, let the free group act on its Cayley tree. Writing an element as a conjugate of a cyclically reduced word exhibits its translation axis, which is unique. A commuting element preserves this axis and acts on it by a translation, not a reflection. Signed translation distance embeds the centralizer into $\mathbb{Z}$: an element with zero translation fixes a vertex, and the left action of a free group on its Cayley tree is free on vertices. The centralizer is therefore cyclic.

Write this centralizer as $\langle b\rangle$, with $t = b^k$, $k \neq 0$. Since $\theta(t)$ has infinite order, so does $\theta(b)$. Hence θ is injective on $\langle b\rangle$, proving the claim. □

6 Verification for the explicit model curve

6.1 Elementary twist conventions

We will use three elementary facts:

$$T_{h(a)} = hT_ah^{-1},$$

twists about disjoint curves commute, and a twist about the boundary of a disk containing exactly two marked points is represented by moving one of those points once around the other.

For completeness, the first two facts follow by transporting, or choosing disjoint, annular supports. For the third, identify the disk with a round disk, place the fixed puncture at its center, and use

$$z \longmapsto e^{2\pi i t\chi(|z|)}z,$$

where χ is one on a disk containing the moving puncture and zero near the boundary. The puncture travels once around the center, and the endpoint is one boundary Dehn twist, with sign determined by the chosen conventions. Reversing the loop gives the other sign. These are also the standard twist and point-pushing facts in [1, Sections 3.3 and 4.2].

Let

$$\eta = \{z : |z| = 2\}, \qquad u = T_\delta, \qquad g = T_{\gamma_*} = uT_Cu^{-1}.$$

The disk bounded by η lies strictly inside the disk bounded by C, and the neighborhood defining δ is disjoint from that disk. Therefore

$$\gamma_* \cap \eta = \varnothing \tag{14}$$

for suitable representatives.

6.2 The monodromy of the required twist

Lemma 6.1. *The element $g = T_{\gamma_*}$ belongs to L and satisfies*

$$(p_x)_*(g) = 1, \qquad (p_y)_*(g) = 1, \qquad |\lambda(j_*g)| = 1. \tag{15}$$

In particular,

$$g(\mathcal{D}_{F_R}) = \mathcal{D}_{F_R}.$$

Proof. Keep $y = R$ fixed and consider the rational map

$$x \longmapsto Z(x, R).$$

By (10), its branch values are $0, 1, R$. Each has two simple ramification points. Its four poles

$$0, \quad 1, \quad 1 - R, \quad \infty$$

are simple.

The Möbius involution

$$J(x) = \frac{(1 - R)(x - 1)}{x + R - 1} \tag{16}$$

satisfies

$$J^2 = \mathrm{id}, \qquad Z(J(x), R) = Z(x, R). \tag{17}$$

Both identities follow by substitution. Its fixed-point equation is

$$x^2 + 2(R - 1)x + 1 - R = 0,$$

which is exactly $Q_y(x, R) = 0$. Thus its fixed points are the two ramification points over R.

It follows that the sheet monodromy of a based meridian about the critical value R is J. Indeed, close to either of these ramification points a meridian interchanges the two local inverse branches, as does J. Transport to the basepoint commutes with J by (17). The assertion holds for either orientation of the meridian.

By the point-pushing description, $u = T_\delta$ is represented by such a meridian about $z = R$, with $y = R$ fixed. Consequently its lift to the four-sheeted covering starts at $x = R$ and ends at $x = J(R)$.

The twist T_C is represented by a meridian about $z = \infty$. Its access path from v_* to infinity may be chosen along the negative real axis, since that ray lies in the complementary disk of C containing v_* and ∞.

Starting on the sheet $x = R$, the lift of this access path tends to $x = \infty$. To see this directly, for real $x \geq R$ all three quantities $Q_0(x, R), Q_1(x, R), Q_y(x, R)$ are positive. Equation (10) therefore gives

$$\frac{d}{dx} Z(x, R) < 0.$$

Moreover,

$$Z(R, R) = v_*, \qquad Z(x, R) \longrightarrow -\infty \quad (x \longrightarrow +\infty).$$

Thus this real inverse branch parametrizes precisely the required lift.

Starting instead at $J(R)$, the lifted access path is its image under J, and therefore tends to

$$J(\infty) = 1 - R.$$

Because this is a simple pole, the infinity meridian lifts to a single meridian about $x = 1 - R$, and returns to the same sheet.

Now represent

$$g = u T_C u^{-1}$$

by the corresponding conjugated configuration loop. Its lift first changes the sheet from R to $J(R)$, then makes the just-described meridian about $x = 1 - R$, and finally returns along the inverse conjugating path. Hence the lift is closed, proving $g \in L$. In the fiber $y = R$, its homotopy class is a based conjugate of a single meridian about $x = 1 - R$.

Since $1 - R \notin \{0, 1\}$, that meridian is null-homotopic after projection to the x-coordinate space M. Its y-coordinate is constant. On the other hand, $x + y - 1 = x + R - 1$ winds once, with one of the two signs, about zero. This proves (15).

The final assertion follows from Lemma 4.3(3). □

6.3 A commuting twist that does preserve the base component

> **Reviewer 2**
>
> Some motivation for considering $t(s)$ would be appropriate here.

Lemma 6.2. *The diagonal loop*

$$t(s) = \big(Re^{2\pi is}, Re^{2\pi is}\big), \qquad 0 \le s \le 1, \tag{18}$$

defines an element $t \in K$ whose mapping class is T_η or T_η^{-1}. Moreover, $(p_y)_*(j_*t)$ has infinite order in $\pi_1(M, R)$.

Proof. The defining factors of (3) do not vanish on $|r| = R = 100$, by the triangle inequality. Thus (18) is a loop in $d(\mathcal{P})$, so $t \in K$.

We identify its mapping class. A calculation from (2) gives

$$\frac{\nu(r)}{r} + \frac{1}{8} = \frac{9r^3 - 21r^2 + 12r - 2}{8r^2(r-1)(2r-1)}.$$

Consequently, when $|r| = R = 100$,

$$\left|\frac{\nu(r)}{r} + \frac{1}{8}\right| \le \frac{9R^3 + 21R^2 + 12R + 2}{8R^2(R-1)(2R-1)} < \frac{1}{100}. \tag{19}$$

Set $a_0 = \nu(R)/R$. The projected configuration loop

$$\big(r(s), \nu(r(s))\big), \qquad r(s) = Re^{2\pi is},$$

has the based homotopy

$$\left(r(s),\ r(s)\left((1-a)\frac{\nu(r(s))}{r(s)} + aa_0\right)\right), \qquad 0 \le a \le 1.$$

Indeed, the expression in parentheses stays in the convex disk of radius $1/100$ about $-1/8$. The moving critical value therefore has modulus greater than 11.5, and is never 0, 1, or $r(s)$. At the end of the homotopy the loop is

$$\big(Re^{2\pi is}, v_* e^{2\pi is}\big).$$

Choose a smooth radial function χ equal to zero on $[0, 2]$ and equal to one on $[10, \infty)$. The ambient isotopy

$$H_s(z) = e^{2\pi is\chi(|z|)}z, \qquad H_s(\infty) = \infty,$$

realizes this last configuration loop. Its endpoint is one Dehn twist in the unmarked annulus $2 < |z| < 10$, and hence is a twist about η. Passing to the deck-action convention may invert it, which proves the asserted description of t.

Finally, the y-coordinate of (18) winds once about zero. Its image in $\pi_1(M, R)$ therefore has infinite order. □

6.4 Infinitely many different components

Proposition 6.3. *The components*

$$g^n(\mathcal{C}_*), \qquad n \in \mathbb{Z},$$

are pairwise distinct.

Proof. By Lemma 6.1, g preserves the deformation space. Suppose that

$$g^n(\mathcal{C}_*) = \mathcal{C}_*$$

for some nonzero integer n. Lemma 4.3(2) gives $g^n \in K$, and hence, by Lemma 5.1,

$$j_*(g^n) \in H.$$

Also $j_*t \in H$.

The curves γ_* and η are disjoint by (14). Thus g commutes with the mapping class of t. Since q_* is injective, they commute in L, and their images commute in $\pi_1(X, w_*)$.

Apply Lemma 5.2 to the free group H and the homomorphism

$$(p_y)_*|_H : H \longrightarrow \pi_1(M, R).$$

By Lemma 6.2, the image of j_*t has infinite order. By Lemma 6.1, the image of $j_*(g^n)$ is the identity. The centralizer criterion therefore gives

$$j_*(g^n) = 1.$$

But (15) and (13) imply

$$\big|\lambda\big(j_*(g^n)\big)\big| = |n| > 0,$$

a contradiction.

Thus no nonzero power of g stabilizes $\mathcal{C}_*$. If two of the displayed components were equal, their index difference would give such a power. They are therefore pairwise distinct. □

7 Transport and completion of the proof

> **Reviewer 2**
>
> All statements in this section are quite straightforward; this can be shortened substantially.

We first justify the isotopy required in Construction 3.3. Along the compact parameter path, the five labelled points

$$0, \quad 1, \quad \infty, \quad r(t), \quad \nu(r(t))$$

remain distinct. On each smooth portion of the path, choose disjoint small disks around the two moving finite points, avoiding 0 and 1. A compactly supported time-dependent vector field which equals the velocity of each moving point near that point has a flow tracking both points and fixing $0, 1, \infty$. Concatenating these flows constructs h_t.

If h_t and $\widetilde{h}_t$ are two such extensions of the same configuration path, then

$$\widetilde{h}_t^{-1} h_t$$

fixes the initial labelled configuration pointwise for every t. Thus the two endpoint pullbacks of γ_* are isotopic relative to that configuration. This proves that the construction specifies an isotopy class without any further choice of ambient isotopy.

Lemma 7.1. *For the isotopy h_t of Construction 3.3, put $h = h_1$. There is a homeomorphism*

$$\Theta : \mathrm{Teich}(B_*) \longrightarrow \mathrm{Teich}(B_{r_0}), \qquad \Theta([\chi]) = [\chi \circ h],$$

such that

$$\Theta(\mathcal{D}_{F_R}) = \mathcal{D}_{F_{r_0}}.$$

It takes $\mathcal{C}_*$ onto the component containing the standard complex structure for F_{r_0}, and intertwines T_{γ_*} with $T_{h^{-1}(\gamma_*)}$.

Proof. Consider the path

$$\tau(t) = [h_t] \in \mathrm{Teich}(B_{r_0}).$$

Its projection to Y is

$$\big(r(t), \nu(r(t))\big).$$

The diagonal path $(r(t), r(t))$ is a lift of this path to W. By Lemma 4.2 and uniqueness of lifting,

$$U_{r_0}(\tau(t)) = (r(t), r(t)).$$

The two paths $F_{r_0}^*(\tau(t))$ and $i_*(\tau(t))$ are therefore lifts to $\mathrm{Teich}(A_{r_0})$ of the same path $r(t)$ in M. They start at the same point, so they coincide for every t. Consequently

$$\tau(t) \in \mathcal{D}_{F_{r_0}}.$$

At $t = 1$, normalized uniformization of the pullback gives a marking h' such that

$$F_R \circ h' = h \circ F_{r_0}. \tag{20}$$

The rational map on the left is F_R because its Hurwitz coordinates are (R, R), and (9) uniquely determines that normalized map. Since $\tau(1)$ lies in the deformation space,

$$h' \simeq h \quad \text{relative to } A_{r_0}.$$

Let

$$\Theta_A : \mathrm{Teich}(A_*) \longrightarrow \mathrm{Teich}(A_{r_0})$$

be change of marking by $h|_{A_{r_0}}$. Naturality of pullback applied to (20), and the relative isotopy of h' and h, give

$$F_{r_0}^* \Theta = \Theta_A F_R^*, \qquad i_* \Theta = \Theta_A i_*.$$

Since Θ_A is a homeomorphism, these identities imply

$$\Theta(\mathcal{D}_{F_R}) = \mathcal{D}_{F_{r_0}}.$$

Furthermore, $\Theta(\tau_*) = [h] = \tau(1)$ lies in the same component as the standard structure $\tau(0)$. Hence Θ carries $\mathcal{C}_*$ onto that component.

Finally, change of marking conjugates a mapping class a to $h^{-1}ah$. Since conjugating a Dehn twist gives the Dehn twist about the inverse-image curve, Θ intertwines the stated twists. □

Proof of Theorem 3.4. Lemma 6.1 and Proposition 6.3 prove the theorem for F_R and γ_*. Lemma 7.1 transfers both preservation of the deformation space and pairwise distinctness of the component orbit to F_{r_0} and $h^{-1}(\gamma_*)$. Undoing the Möbius normalization transfers these conclusions to f and the curve (6). □

Remark 7.2. The proof distinguishes two assertions which should not be conflated. The vanishing of both coordinate monodromies proves that the twist preserves the *whole* deformation space. Its failure to preserve any nonzero-power iterate of the base component is a separate fact, detected by the winding homomorphism (13) together with a cyclic-centralizer argument in the free group H.

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